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Cooling Tower

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**BALTIMORE
AIRCOIL COMPANY**

Technical Datasheet Cooling Tower



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Model Information

Product line Compass Cooling Tower
Model CPSC-0817-08L

Main Accessories

Intake option Combined Inlet Shields (CIS) - Replace Air Inlet Louvers

Thermal Design Data

Requested capacity (unit) 5,077.97 MBH
Maximum capacity (unit) 5,095.61 MBH
Reserve capacity 0.14 %
Requested fluid flow 1,016.00 gpm
Entering wet bulb temperature 70.00 °F
Inlet temperature of process fluid 88.00 °F
Outlet temperature of process fluid 78.00 °F

Physical Data (per unit)

Overall length (including all options and accessories) 8' 10"
Overall width (including all options and accessories) 17' 5"
Overall height (including all options or accessories) 13' 6"
Shipping weight 5,639 lbs
Operating weight 13,757 lbs
Heaviest section 0 lbs
Material option Galvanized Steel with FRP Hot & Cold Water Basin
* Note: Exact unit dimensions and weights may be influenced by accessories/option combinations.

Wet Deck Surface

Wet deck surface material PVC

Electrical Data (per unit)

Fan motor
BHP (net mech output power at motor shaft) 15.0 HP
Voltage 200 V
Frequency 60 Hz
Protection class IP55
Fan motor power (1) 15.0 = 15.0 HP/Unit
Efficiency class IE3 Inverter Duty
RPM 1,800
Type 1 Speed/1 Winding
Enclosure TEAO : Totally Enclosed Air Over/Airtight Motor

Aerodynamic Data (per unit)

Airflow (100% RPM/100% RPM) 90,476 CFM
Fan type Standard Fan

Hydraulic Data (per unit)

Total pressure drop 5.66 PSI

Special Requirement(s)

Notes:

- Thermal performance at design conditions and total standard fan motor power is CTI/ECC certified
- Refer to unit print for detailed dimensions.

Our reference: 62239 100 / 0
Your reference: LO PROYECTO AYURÁ
Print Version: 2024-07-03_17-23

www.BaltimoreAircoil.com



Sound Rating Cooling Tower

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Date: 7/3/2024

Project Name: LO PROYECTO AYURÁ
Project City: Miami
Project Country: United States

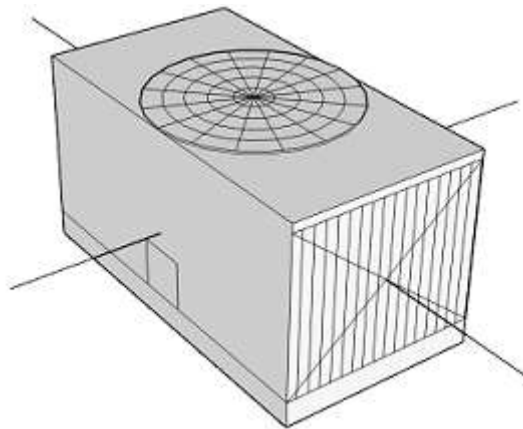
Model Information

Product line: Compass Cooling Tower
Model: CPSC-0817-08L
Number of units: 1
Total pump motor power:
Total standard fan power: 100%, 15.0 HP/Unit
Fan type: Standard Fan
Intake option: Combined Inlet Shields (CIS) - Replace Air Inlet Louvers
Internal option:
Discharge option:
Fan motor: (1) 15.0 = 15.0 HP/Unit

Octave band and A-weighted sound pressure levels (Lp) are expressed in decibels (dB) reference 0.0002 microbar. Sound power levels (Lw) are expressed in decibels (dB) reference one picowatt. Octave band 1 has a center frequency of 63 Hertz.

Air Inlet Sound Pressure (dB)		
Octave Band	Distance	
	5 ft.	50 ft.
1	82	65
2	80	64
3	77	60
4	73	59
5	63	52
6	59	52
7	55	50
8	47	45
A-wgtd	74	60

End Sound Pressure (dB)		
Octave Band	Distance	
	5 ft.	50 ft.
1	73	67
2	73	62
3	69	57
4	63	55
5	54	49
6	53	41
7	50	38
8	40	37
A-wgtd	65	56



Top Sound Pressure (dB)		
Octave Band	Distance	
	5 ft.	50 ft.
1	83	72
2	83	72
3	80	70
4	75	62
5	66	53
6	63	51
7	59	47
8	49	37
A-wgtd	76	65

End Sound Pressure (dB)		
Octave Band	Distance	
	5 ft.	50 ft.
1	73	67
2	73	62
3	69	57
4	63	55
5	54	49
6	53	41
7	50	38
8	40	37
A-wgtd	65	56

Air Inlet Sound Pressure (dB)		
Octave Band	Distance	
	5 ft.	50 ft.
1	82	65
2	80	64
3	77	60
4	73	59
5	63	52
6	59	52
7	55	50
8	47	45
A-wgtd	74	60

Sound Power (dB)		
Octave Band	Center Frequency (Hertz)	Lw
1	63	104
2	125	104
3	250	102
4	500	94
5	1000	86
6	2000	83
7	4000	80
8	8000	72
A-wgtd		97

Notes

- The use of VFDs can increase sound levels.
- The values presented in this document are estimates only and are subject to change by BAC.
- Sound rating is for a single unit.
- Sound data provided by 'ATC 128 (2019)' sound test code.
- Sound data is free field data valid for unit installation without elevation, not taking into account any reflections. Octave band values are shown for indicative purposes only. Values are obtained according to CTI ATC-128 (Test Code for Measurement of Sound From Water-Cooling Towers) for small towers.



Mechanical Specifications

Cooling Tower

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Background Information

Customer:	SAEG ENGINEERING GROUP-FL		
Project:	LO PROYECTO AYURÁ		
BAC Quote Number:	62239	Line:	100
Revision:	0	Revision Name:	MASTER
Purchase Order No:			
Engineer:			

Model Information

Product line:	Compass Cooling Tower
Model:	CPSC-0817-08L
Quantity:	1
Knockdown For Field Assembly?	Yes

Mechanical Specifications

Unit Type	Induced Draft;; Crossflow Cooling Tower Induced draft;; crossflow cooling tower with vertical discharge. The unit is supplied disassembled (knocked-down) for field assembly. The unit is not capable for lifting.
Quality Assurance	Quality Assurance - ISO 9001 Compliant Each unit is manufactured under closely-controlled conditions using standardized parts to ensure each unit is built precisely to the same high-quality design and construction standards. The design, manufacture, and business processes of Baltimore Aircoil Company are ISO 9001 compliant.
Unit Efficiency	Unit Energy Efficiency per ASHRAE Standard 90.1-2013 The unit(s) complies with the energy efficiency requirements established by ASHRAE Standard 90.1-2013.
CTI Certification	CTI Certified Thermal Performance The thermal performance of this BAC unit has been certified through performance tests conducted by the Cooling Technology Institute in accordance with their standard STD-201 RS. Your equipment may be selected for factory-testing to verify CTI certified performance. Such certification by an independent third party assures engineers and users that the published thermal capacities accurately reflect the actual unit performance. CTI certification eliminates the additional costs of on-site, individual unit testing, oversizing the equipment or operating cost penalties from deficient equipment.
Materials of Construction	Structural Elements are Constructed of Heavy Gauge Galvanized Steel All structural steel components are constructed from G-235 (Z700 metric) hot-dip galvanized steel. The edges of the hot-dip galvanized steel components are given a protective coat of zinc-rich compound. Hot water distribution basins are gravity type constructed of heavy gauge, G-235 (Z700 metric) hot-dip galvanized steel. Polypropylene metering orifices are provided to assure even distribution of water over the wet deck surface. Cold water basin (CWB) and Fan cowl are non-corroding, fiberglass-reinforced polyester (FRP). CWB includes a depressed section with drain/clean-out connection and the area under the fill sections is sloped toward the depressed section for easy cleaning.
HWB and CWB	GLV Hot Water Basin and FRP Cold Water Basin The hot water distribution basins are open gravity type for easy cleaning, and constructed of hot-dip galvanized steel G235 (Z700). Gravity flow nozzles are snap-in type for easy removal. The cold water basin is made of FRP that is corrosion resistant. The cold water basin has a depressed center section with drain/clean-out connection. The Basin area under the fill is sloped toward the depressed center section to facilitate cleaning. Standard cold water basin include a copper make-up valve that installed for operating water level.

Hot Water Basin Cover	FRP Hot Water Basin Cover FRP HWB covers are furnished to prevent the accumulation of debris and algae in the hot water distribution basins. They can be served as service platform if necessary.
Fan & Drive System	Standard Fan Driven by Sheave & Belts Fan(s) are driven by multiple neoprene/polyester V belt designed specifically for evaporative cooling equipment service. Motor is mounted on an adjustable motor base. Fan sheaves is non-corrosive cast aluminum, motor sheave is cast Iron. Fan(s) and steel fan shaft(s) are supported by heavy-duty, self-aligning, grease-packed, relubricatable ball bearings with special seals for protection against dust and moisture. All bearings are designed for minimum L10 life of 30,000 hours. Extended lubrication lines are optional.
Motor Space Heater	Motor Space Heater All induction motors for fans are equipped with space heater as standard. Space heater will prevent condensation from generating and accumulating inside the motor to further protect the motor. Unless otherwise stated, voltage of space heaters is 110-120 VAC/ 50 or 60 Hz based on motor configuration. If the other requirement is needed, please contact your local BAC distributor. The power of the motor and the space heater should be interlocked. Make sure that the space heater is switched off before the motor is switched on and turned on two hours after the motor is turned off. Please refer to the space heater wiring diagram included in the submittal or on the motor.
Fan Guard(s)	Galvanized Steel Fan Guard A heavy gauge, hot-dip galvanized steel wire fan guard is provided over each fan cylinder. The fan guard is shipped loose for field installation.
Casing Panels	Fiberglass Reinforced Polyester (FRP) Casing Panels Casing panels are constructed of corrosion resistant, UV protected Fiberglass Reinforced Polyester (FRP).
Fill	PVC Fill & Drift Eliminators The BACross® Fill and integral drift eliminators are formed from self-extinguishing (per ASTM D-568) polyvinyl chloride (PVC), having a flame spread rating of 5 per ASTM Standard E84-77a, and are impervious to rot, decay, and fungus or biological attack. The fill is elevated above the cold water basin floor to facilitate cleaning. This fill is suitable for a maximum entering water temperature of 130°F (54.44°C). The eliminators are designed to effectively strip entrained moisture from the leaving airstream with a minimum of air resistance.
Water Inlet(s)	Top Inlet Connections Two (2) hot water inlet flange pattern connections, suitable for ISO7005 PN1.0Mpa flat face flanges, are located at the top of the designated cell(s).
Water Outlet(s)	Bottom Outlet Pump Suction Connection The unit is provided with a bottom bolt hole pattern(s) for mating with an ISO7005 PN1.0Mpa flat face flange. The bolt hole pattern is located in the depressed area of the cold water basin and appropriately sized for design flow. The flat face flange, full face gasket, and hardware are supplied and installed by others. Also included is a large area, lift out strainer made of SST304 has perforated openings sized smaller than the water distribution nozzle orifices. Strainer includes anti-vortexing baffle to prevent air entrainment. Please see the submittal package for the connection size.
Basin Water Level Control	Mechanical Float Valves for Independent Cell Operation Each cell is provided with one make-up valve with unsinkable polystyrene filled plastic float arranged for easy adjustment. The make-up valve is suitable for water supply pressures between 15 psig (103 kPa) and 50 psig (345 kPa).
Air Intake Option	Combined Air Inlet Shields The air intake face of the unit is protected by a corrosion and UV resistant PVC shield. This shield prevents airborne debris from entering the unit(s).
Access Door	Standard Access Door Standard Access Door on (1) side for single cell or (2) sides for multi-cell.
Internal Access Option	GLV Internal Walkway A G-235 (Z700 metric) galvanized steel walkway complying with OSHA standards and regulations provides access to the plenum to facilitate servicing the unit.
Crating/ShippingType	Knocked Down Unit The unit will be shipped knocked down in minimum crating with instructions for field assembly. Field assembly is provided by others. Due to the limitation of chemicals transportation, the resin is also provided by others.



Submittal Data

Cooling Tower

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Background Information

Customer:	SAEG ENGINEERING GROUP-FL		
Project:	LO PROYECTO AYURÁ		
BAC Quote Number:	62239	Line:	100
Revision:	0	Revision Name:	MASTER
Purchase Order No:			
Representative:	Luisa Orozco (LA - Other (CN-SH)) ☒ luisa.orozco@saeg.com		

Model Information

Product line:	Compass Cooling Tower		
Model:	CPSC-0817-08L		
Quantity:	1		
Certified Capacity Per Unit:	Maximum capacity (unit):	5,095.61 MBH	
	Water Flow Rate per Unit (GPM):	1,016.00 gpm	
	Entering Water Temp (°F):	88.00 °F	
	Entering Wet-Bulb Temp (°F):	70.00 °F	
	Leaving Water Temperature (°F):	78.00 °F	
Fan Motor(s):	Enclosure:	TEAO : Totally Enclosed Air Over/Airtight Motor	
	Fan Drive System:	BALTIDRIVE® Power Train	
	Total standard fan power:	100%, 15.0 HP/Unit	
	Fan motor:	(1) 15.0 = 15.0 HP/Unit	
	System Voltage:	200 V	
	System Frequency:	60 Hz	
	System Phase:	3 Phase	
	Fan Motor Type:	1 Speed/1 Winding	
	Horsepower Motor A:	15.0 HP	
	Fan Motor Options A:	Space Heater	

Execution:

Notes:

Drawings And Diagrams

Drawing Type	Drawing Number	Drawing Type	Drawing Number
Unit Print	UP-62239-100	Bottom Connections.....	BC-62239-100
Unit Support	SS-62239-100		



Unit Print

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HWB COVER
(OPTIONAL, SEE NOTE)
热水盘盖, 可选项,
见技术要求

DN150 INLET
DN150 进水孔

/U6

2698

1349

5307

ACCESS DOOR
安全门

MODEL

SHIP WEIGHT

OPER WEIGHT

"H"

"A"

型号

运输重量(KG)

运行重量(KG)

尺寸(mm)

尺寸(mm)

CPSC-0817-07J	2232	5753	3716	3563
CPSC-0817-07K	2243	5765		
CPSC-0817-07L	2289	5810		
CPSC-0817-07M	2312	5833		
CPSC-0817-07N	2351	5873		
CPSC-0817-07O	2376	5898	4123	3966
CPSC-0817-08J	2367	5888		
CPSC-0817-08K	2378	5900		
CPSC-0817-08L	2424	5946		
CPSC-0817-08M	2447	5968		
CPSC-0817-08N	2486	6008		
CPSC-0817-08O	2511	6033		
CPSC-0817-08P	2545	6067		

Notes:

1) Hot water basin cover is optional and not included in standard configure.

2) Drawings are not to scale. ALL dimensions are in "mm".

Weights are in kg and exclude options and accessories.

3) Field piping should be fabricated at time of installation. Pre-fabrication of pipe work is not recommended

4) Do not support piping from unit connections. All necessary piping support to be supplied by others within 500mm distance from connection flange.

5) The area above the fan discharge must be unobstructed.

6) Connections 4" and larger are furnished with one stud circle, suit for PN1.0 MPa per China GB/T 9119 or ISO 7005.1.

注意事项:

1) 热水盘盖是额外选项, 不包含在标准配置中;

2) 图纸不是按比例的. 所有尺寸单位是 "mm", 重量单位是 "kg" (不包括选配件和附件);

3) 在塔安装的时候进行现场配管, 不推荐预安装管道的工

作;

4) 不要在连接处提供管路支撑, 在距离连接法兰500mm处提供管路支撑;

5) 风扇出口风口处不允许有障碍物.

6) 大于等于4"的孔适用于平面法兰PN1.0MPa(GB/T 9119 或ISO 7005.1)

QUOTE NO: 62239-100

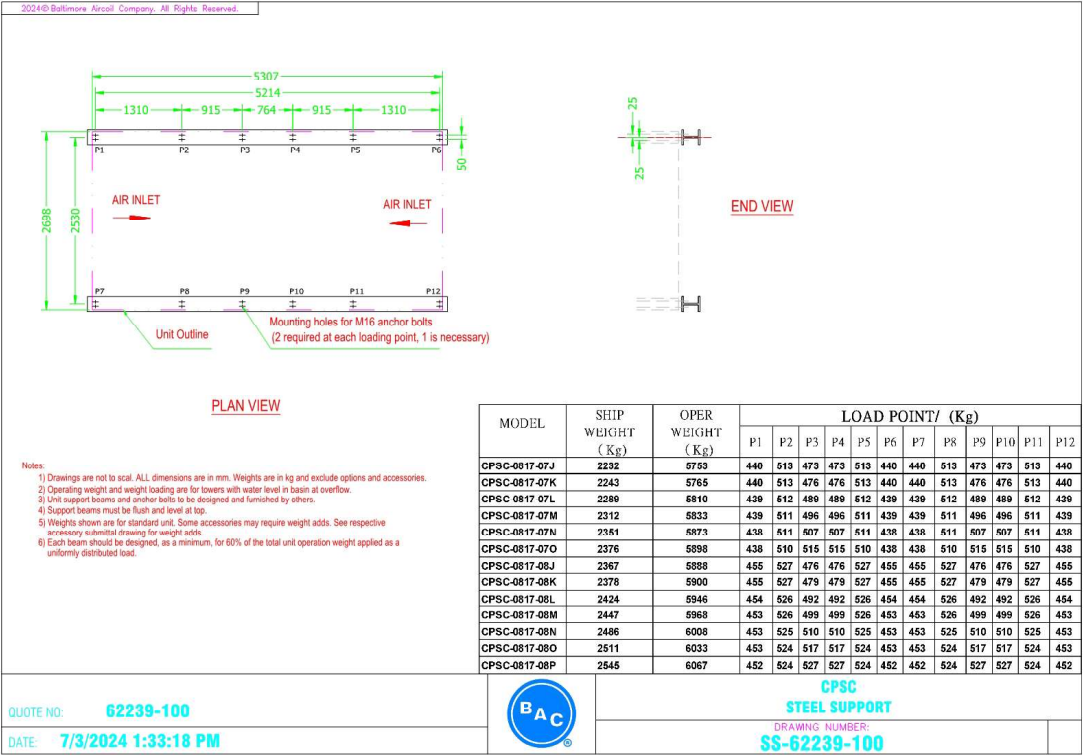
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CPSC UNIT PRINT

DRAWING NUMBER: UP-62239-100



Unit Support





Bottom Connections

